

Constructing Inpatient Pressure Injury Prediction Models Using Machine Learning Techniques

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The incidence rate of pressure injury is a critical nursing quality indicator in clinic care; consequently, factors causing pressure injury are diverse and complex. The early prevention of pressure injury and monitoring of these complex high-risk factors are critical to reduce the patients' pain, prevent further surgical treatment, avoid prolonged hospital stay, decrease the risk of wound infection, and lower associated medical costs and expenses. Although a number of risk assessment scales of pressure injury have been adopted in various countries, their criteria are set for specific populations, which may not be suitable for the medical care systems of other countries. This study constructs three prediction models of inpatient pressure injury using machine learning techniques, including decision tree, logistic regression, and random forest. A total of 11 838 inpatient records were collected, and 30 sets of training samples were adopted for data analysis in the experiment. The experimental results and evaluations of the models suggest that the prediction model built using random forest had most favorable classification performance of 0.845. The critical risk factors for pressure injury identified in this study were skin integrity, systolic blood pressure, expression ability, capillary refill time, and level of consciousness.

KEY WORDS: Classification technique, Inpatient pressure injury, Machine learning, Pressure injury risk assessment

Inpatient pressure injury is a critical clinical monitoring indicator; a high rate of inpatient pressure injury not only indicates many underlying medical problems but also

predicts potential subsequent problems. Pressure injury is strongly associated with prolonged hospital stays, and may be accompanied by wound infection, pain, numerous complications, and even death if severe.¹ Thus, pressure injury is a key factor for medical care quality. Ninety-five percent of pressure injury cases can be prevented, and the cost of preventing pressure injury is much lower than the cost of treatment.² A previous study conducted in the United States revealed that the medical expenses of a patient with pressure injury range from US\$2000 to US\$3000 on average.³

Pressure injury is preventable; however, identifying all risk factors is difficult due to complex and diverse causes.⁴ The Braden Scale⁵ is the most commonly used clinical tool for pressure injury prediction as a result of its high sensitivity and specificity. However, previous studies have shown that some risk factors for pressure injury, such as sex, length of hospital stay, disease severity, medication use, diagnosis, and nursing workload, are not taken into consideration within the Braden Scale.⁵

The pressure injury assessment scales currently used by hospitals in Taiwan for the systematic checkup of patients prior to hospital admission are derived from the pressure injury risk assessment scales of other countries. To date, there has been no single universal pressure injury assessment scale developed for Taiwan. The Joint Commission of Taiwan provides hospitals with various healthcare quality indicators and training programs to improve healthcare for patients. The Taiwan Clinical Performance Indicator (TCPI) evaluates clinical care in four categories, including general care, acute psychiatric care, chronic psychiatric care, and long-term care. The prevalence rate and density of pressure injury among patients are used to monitor the quality of hospitalization care. The incidence rate of inpatient pressure injury in Taiwan's regional hospitals was 0.22% in 2015 according to the strict reporting requirements of TCPI.⁶ Despite compliance with these scales, the occurrence rate of pressure injury has remained the same in Taiwan.⁵ This suggests that some risk factors are not included in the clinical pressure injury assessment scales. Furthermore, the prediction power of any single assessment mentioned above is weak, possibly due to the complexity and interdependent nature of the risk factors for pressure injury. Therefore, the establishment of an accurate pressure injury risk assessment scale is urgently required.

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Because of the nursing human-resource shortage and heavy workload, a comprehensive pressure injury assessment scale is needed to assist nurses in identifying the actual risk factors for pressure injury and preventing it in high-risk patients. After identification, preventive care can be provided to patients to reduce the incidence rate of pressure injury and prevent advancement of the pressure injury stage.^{7,8}

Raft⁹ conducted a study using logistic regression (LGR) and found the following risk factors for pressure injury: spinal cord injury, mechanical ventilation, and age of 40 years or older. Some demographic information such as age, hospital department, and sex were included in many previous studies.^{10–16}

Raju et al¹⁷ used LGR, decision tree (DT), and random forest (RF) to examine risk factors for general inpatient pressure injury and ranked them in the following order according to their level of significance: length of hospital stay, serum albumin, age, blood urine nitrogen, and total Braden score. Lyder et al¹⁸ analyzed retrospective data on pressure injury and showed that age, cancer, congestive heart failure, chronic obstructive pulmonary disease, cardiovascular diseases, diabetes, obesity, rehospitalization, and prolonged length of hospital stay were all significantly correlated with pressure injury. Using multivariate linear regression, Cox¹⁹ and Cremasco et al²⁰ analyzed the development of pressure injury in patients in the intensive care unit and identified the following risk factors for pressure injury: sex, age, length of hospital stay, mobility, frictional/shear forces, adrenaline infusion, cardiovascular diseases, disease severity, and nursing workload. In addition to Cox,¹⁹ the length of hospital stay and medical history^{11,14,15,18,21} and the level of mobility^{5,10,11,22,23} were also commonly considered.

Lahmann et al^{24,25} used the χ^2 automatic interaction detection to model a classification tree for identifying the predictors of pressure injury and showed that friction and shear are the most potent predictors of pressure injury. Okuwa et al²⁶ used LGR to analyze risk factors for pressure injury in the lower limbs and reported that the key risk factors were length of bed rest, male sex, lower limb contracture, toe contracture, ankle-brachial index, hemoglobin, red blood cells, sodium ions, and chloride ions. Schoonhoven et al²⁷ analyzed the independent predictors of pressure injury by using LGR and found that the predictors were age, body weight at admission, abnormal appearance of the skin, friction and shear, and planned surgery.²⁸ Skin appearance and properties were considered as one major indicator of pressure injury.^{3,4,10,13–16,22,25,29}

This study used classification techniques to establish a domestic inpatient pressure injury prediction model intended to assist nurses to identify patients at high risk. This model automatically screens and identifies patients who are at a potential high risk of pressure injury using the electronic medical record, thereby simplifying the current complex process that

requires repeated assessment of pressure injury risk factors. Another advantage of a pressure injury prediction model is that it can be used for case management; in other words, it enables nurses to implement early preventive treatments for pressure injury, which may reduce related complications and eventually improve medical care quality.

MATERIALS

Data Collection

This study was conducted using clinical data from one regional teaching hospital in Taiwan; it is a metropolitan hospital with 455 beds. The study applied for access to patients' electronic medical records and related information from the institutional review board of the hospital. The data were collected from patients hospitalized between the years 2014 and 2016. Each admission of a patient at a different time was considered as a single data point and all data were deidentified in the analysis. Patients' medical information, including nursing assessment at admission, laboratory test results, and pressure injury-related information, were collected retrospectively. When the data values were missing, the authors performed value imputation based on the similarity between samples and deleted samples if too many values were missing; an excess of missing values may lead to analysis errors. The literature shows that the causes of pressure injury differ significantly between adults and children.³⁰ Therefore, the present study targeted adults and excluded patients younger than 18 years.

Variables

In Taiwan, domestic pressure injury assessment scales are mostly derived from the Braden Scale,⁵ Gosnell Scale,²³ Waterlow Scale,²⁹ and Norton Scale.³¹ These scales are used by different hospitals, and the assessment criteria are set up differently according to the target population as well as applicability and feasibility in terms of medical care systems. The contents of these four pressure injury assessments are different, and the criteria for determining the risk of pressure injury vary. To facilitate clinical use, these measurements do not include all variables that may affect pressure injury; therefore, using these measurements is not enough.

According to previous studies, risk factors for pressure injury can be divided into internal and external factors. The internal factors include patients' health problems or conditions, which are sex, age, malnutrition, albumin level, anemia, hypotension, incontinence, decreased level of consciousness in daily activities, systemic infection, chronic diseases, disease severity, smoking habit, increases in physical and mental stress, and negative emotions. The external factors are the amount of pressure imposed on tissues, length of stay, shear force, friction, and high humidity.^{1,21,32,33}

The Barthel Index represents the activities of daily living assessment scale, which is the most commonly used scale to assess physical function in the case of long-term care and home care of the National Health Insurance. The scoring items in the Barthel Index include the ability to feed, move from wheelchair to bed and return, personal toilet, get on and off toilet, bathe self, walk on level surface, ascend and descend stairs, dress, and control bowels and bladder. A higher total score on the Barthel Index means the patient is less dependent on assistance. The total score on the Barthel Index and the incidence of pressure injury were negatively correlated; this implies a higher likelihood of patients being bedridden, which in turn can increase the rate of pressure injury because of prolonged pressure on the skin.

After integrating information from previous literature, existing scales, and clinical recommendations, this study selected 54 variables under 12 categories (Table 1).

DATA ANALYSIS

After data collection and preprocessing, a total of 11 677 non-pressure injury cases and 161 pressure injury cases were included in the data analysis. The data set was randomly split into training and validation sets (1:1 ratio). That is, 50% of the data were used to train the best classifier in the tests and the remaining 50% of the data were used to validate the model performance. To select an optimal feature set that contains the independent variables highly correlated with the dependent variable (class) but uncorrelated with each other, a correlation-based feature selection technique, the CfsSubsetEval module of the data mining software Waikato Environment for Knowledge Analysis (WEKA; University of Waikato, New Zealand), was applied.

In the collected dataset, the rate of pressure injury was 1.36%, suggesting an imbalanced ratio between the two class labels. To avoid the class imbalance problem, the resample module of WEKA was employed to both undersample the majority class and oversample the minority class by the bootstrap method. As a result, the injury and non-pressure injury cases were adjusted at a 1:1 ratio in the training set. Moreover, some useful samples in both classes may not be chosen by the resample module, resulting in a possible bias in the evaluations of model performance. Therefore, the bootstrap method was applied 10 times to construct 10 training datasets; for each generated training dataset, 10-fold cross-validation was then conducted in all the experimental evaluations.³⁵

In this study, WEKA was used to build prediction models of DT, LGR, and RF. Logistic regression is similar to the linear regression analysis that mainly explores the relationship between the dependent variable and the independent variable. The dependent variable in linear regression is usually a continuous variable; however, in this case, the dependent variable discussed by LGR is mainly a categorical variable,

especially a variable divided into two categories. In addition to using a binary dependent variable, there are other extensions for more than two levels of dependent variables, including multinomial logistic regression or ordinal LGR. Decision tree uses the concept of tree branching as a decision-making model, which can be applied to classification and regression. Because the prediction result can display not only the category type but also the numerical data, the DT can be expanded into the classification tree, regression tree, or classification and regression tree. Random forest can be seen as an ensemble learning model of multiple DTs. Similar to the DT, the RF can be applied to classification and regression. When a new target is classified by RF, each tree in the RF will give its own classification label and vote for it. The overall output of the forest will be the classification option with the most votes and the average of all DT outputs in the regression problem. The DT, the LGR, and the RF models were constructed using the J48, SimpleLogistic, and RandomForest modules in WEKA, respectively. Because the prediction performance of classifiers can be significantly influenced by the hyperparameter settings, the CVPParameterSelection meta-learner module of WEKA was used to optimize the hyperparameters for each classifier. Table 2 shows the details of the parameter tuning for each classification technique.

After the best classifier is determined from the test of training sets, the best classifier is then used with optimal parameter settings to examine the performance using validation set.

The confusion matrix was adopted to evaluate the performance of each prediction model (Figure 1). True-positive (TP) is the number of inpatient pressure injuries in the sample and predicted to be successful; true-negative (TN) is the number of inpatient non-pressure injuries in the sample and predicted to be inpatient without pressure injury; false-positive (FP) is the number of inpatient pressure injuries that were actually hospitalized for non-pressure injury patients in the sample; and false-negative (FN) is the number of inpatient non-pressure injuries that were actually hospitalized for pressure injury in the sample.

Using the information summarized in the confusion matrix, four classification performance metrics, including the precision, the recall (ie, sensitivity), the specificity, and the F1-measure, can be obtained through the following equations:

$$Precision = \frac{TP}{TP + FP} \quad (1)$$

$$Recall = Sensitivity = \frac{TP}{TP + FN} \quad (2)$$

$$Specificity = \frac{TN}{FP + TN} \quad (3)$$

Table 1. Variables

Category	Variable Name	Type	Definition	Reference
Activities and exercise	Barthel Index total	Cat.	0–20: Total dependence/21–60: severe dependence/61–90: moderate dependence/91–99: slight dependence	9
	Cardiovascular system	Cat.	0: Cardiovascular was normal during the hospitalization/1: cardiovascular was abnormal during the hospitalization	20
	Daily activities	Cat.	0: Daily activities was normal during the hospitalization/1: daily activities was abnormal during the hospitalization	
	Left lower limb	Cat.	Refers to muscle strength 0: No movement/1: flicker of movement only/2: movement possible when assisted by gravity or gravity is eliminated/3: movement possible against gravity but without imposed resistance/4: weak movement possible against gravity/5: normal movement against gravity and against imposed resistance	27
	Left upper limb	Cat.	Refers to muscle strength 0: No movement/1: flicker of movement only/2: movement possible when assisted by gravity or gravity is eliminated/3: movement possible against gravity but without imposed resistance/4: weak movement possible against gravity/5: normal movement against gravity and against imposed resistance	27
	Right lower limb	Cat.	Refers to muscle strength 0: No movement/1: flicker of movement only/2: movement possible when assisted by gravity or gravity is eliminated/3: movement possible against gravity but without imposed resistance/4: weak movement possible against gravity/5: normal movement against gravity and against imposed resistance	27
	Right upper limb	Cat.	Refers to muscle strength 0: No movement/1: flicker of movement only/2: movement possible when assisted by gravity or gravity is eliminated/3: movement possible against gravity but without imposed resistance/4: weak movement possible against gravity/5: normal movement against gravity and against imposed resistance	27
Cognitive sensory pattern	Emotional responses	Cat.	0: Emotional responses was normal during the hospitalization/ 1: emotionally indifferent was normal during the hospitalization/ 2: emotional anxiety was normal during the hospitalization/3: emotional noresponses were normal during the hospitalization	5
	Pain score	Cat.	0: No pain/1: minor/2: moderate/3: severe	1,5
Conscious	Expression ability	Cat.	0: Normal/1: bad	34
	Eye opening (E)	Cat.	1: None/2: to pain/3: to speech/4: spontaneous	34
	Motor response (M)	Cat.	1: No response/2: decerebrate extension/3: decorticate flexion/ 4: withdrawal/5: localize/6: obey	34
	Orientation	Cat.	0: Normal/1: confused	34
	Verbal response (V)	Cat.	1: None/2: unintelligible sounds/3: inappropriate words/4: confused/5: oriented	
Demographic information	Age	Num.	Hospital days minus the date of birth	10–12,14,17,18,20
	Blood type	Cat.	A: type A/B: type B/O: type O/AB: type AB	27
	BMI	Cat.	0: Normal ($8.5 \leq \text{BMI} < 24$)/1: too high ($24 \leq \text{BMI}$)/2: too low ($\text{BMI} < 18.5$)	19,26,35
	Hospital department	Cat.	1: general surgery/2: gastroenterology/3: respiratory care center/4: infectious disease/5: oncology/6: neurology/7: cardiology/8: orthopedics/9: nephrology/10: chest medicine/ 11: endocrinology/12: neurosurgery/13: urology surgery/14: pain management/15: otolaryngology, head and neck surgery/ 16: oral and maxillofacial surgery/17: thoracic surgery/18: obstetrics and gynecology/19: cardiovascular surgery/20: dermatology/21: rheumatology and immunology	13–17,35
	Sex	Cat.	0: Female/1: male	11,12,14,17,21,27

(continues)

Table 1. Variables, Continued

Category	Variable Name	Type	Definition	Reference
Excretion function	Defecation pattern	Cat.	0: The defecation was normal during the hospitalization/1: the defecation was incontinent during the hospitalization/2: the defecation was diarrhea during the hospitalization/3: the defecation was constipation during the hospitalization	
	Total assessment	Num.		
	Voiding pattern	Cat.	0: Urination was normal during the hospitalization/1: urination was abnormal during the hospitalization	
Habits	Alcohol drinking	Cat.	0: No/1: yes	
	Betel nut chewing	Cat.	0: No/1: yes	
	Smoking	Cat.	0: No/1: yes	11
Lab Data	AST	Num.	Aspartate aminotransferase	
	BUN	Num.	Blood urea nitrogen	18
	Creatinine	Num.		
	Glucose-AC	Num.	Glucose-ante cibum	
	Na+	Num.		27
	T-Bilirubin	Num.		
	Uric Acid	Num.		
Medical history	WBC	Num.	White blood cell	18
	Family history of cancer	Cat.	0: No/1: yes	19
	History of allergy	Cat.	0: No/1: yes	
	Long-term use of medications	Cat.	0: No/1: yes	11,20,22
Pressure injury high-risk group assessment	Medical history	Cat.	0: No/1: yes	11,15,16,20,22
	Album level	Cat.	1: Alb $\leq$ 2.5/2: Alb 2.6–3/3: Alb 3.1–3.5/4: Alb $\geq$ 3.6	10
	Circulation	Cat.	1: Medium and severe edema/2: slight edema/3: delayed blood vessels fill/4: microvascular fill too fast	7,29
	Coma Scale	Cat.	1: GCS < 7/2: GCS 8–11/3: GCS 12–14/4: GCS 15	
	Excretion pattern	Cat.	1: Incontinence/2: stool incontinence/3: occasional urinary incontinence/4: control excretion	24,30,32
	Level of activity	Cat.	1: Bedridden/2: wheelchair/3: need assistance when walking/4: independent activities	5,24,30,32
Skin appearance and properties	Level of mobility	Cat.	1: Immobility/2: unable to turn over/3: turn over/4: fully motivated	5,11,14,20,23,24
	Capillary refill time	Cat.	0: Normal/1: abnormal	3,4,13,16,17,26
	Color at the peripheral limbs	Cat.	0: Normal/1: pale/2: redness	3,4
	Skin integrity	Cat.	0: Complete/1: incomplete	3,4,13–17,23
	Skin properties	Cat.	0: Normal/1: edema/2: dry/3: inelastic	13,15,30
Vital signs	Temperature at the peripheral limbs	Cat.	0: Warm/1: cold/2: fever	3,4
	Diastolic pressure	Num.		32
	Pulse rate	Num.		32
	Respiration rate	Num.		32
	Systolic pressure	Num.		32
	Temperature	Num.		32

Abbreviations: Cat., categorical; Num., numerical; BMI, body mass index; AST, aspartate aminotransferase; BUN, blood urea nitrogen; WBC, white blood cell; GCS, Glasgow Coma Scale.

Table 2. Hyperparameter Tuning in WEKA

Technique	Hyperparameters	Range	Increment
DT	Confidence factor	0.1–0.5	0.05
	Minimum of number of instances per leaf	2–20	1
LGR	Ridge value = 1.0E-8	Default	NA
RF	Number of trees	50–250	10

$$F1\text{--measure} = 2 \cdot \frac{\text{Precision} \cdot \text{recall}}{\text{Precision} + \text{recall}} \quad (4)$$

This study also included area under the curve (AUC) as an indicator of the quality of the models. The AUC ranges from 0 to 1, and a higher value denotes the more favorable performance of the classifier.

RESULTS

Statistical Analysis

For statistical significance, categorical variables were tested using the χ^2 test, and continuous variables were tested using the paired t test. The results (Supplemental Digital Content 1, <http://links.lww.com/CIN/A59>) showed that inpatient pressure injury was significantly influenced by 36 of 54 variables, which were as follows: hospital department, age, body mass index, smoking habits, disease history, pulse rate, respiration rate, systolic blood pressure, diastolic blood pressure, eye opening response, verbal response, motor response, expression ability, orientation, sodium ion, temperature of the peripheral limbs, color of the peripheral limbs, skin integrity, capillary refill, skin properties, Coma Scale, excretion pattern, level of mobility, level of activity, circulation, pressure injury high-risk group assessment, voiding pattern, defecation pattern, upper and lower limb scores, daily activities, total score of Barthel Index, emotional response, and pain score (Supplemental Digital Content 1, <http://links.lww.com/CIN/A59>).

Variable Importance

Attribute selection was performed using WEKA; the GainRatioAttributeEval module was used to evaluate the level of importance of the 36 variables that affected inpatient pressure injury. This module computes the gain ratio for each independent variable, which is a common metric used

to measure the variable importance in DT-based classifiers. The gain ratio represents the percentage of information gain of one attribute over all attributes, which could be used to evaluate the importance of an attribute. The results shown in Table 3 revealed that skin integrity was the most important risk factor, followed by systolic blood pressure change, expression ability, the capillary refill time, and the level of consciousness.^{35,36}

Patients with impaired skin integrity were at a high risk of pressure injury. Inpatients with low skin integrity already had wounds when admitted to the hospital, which will increase the complexity of factors causing skin abnormality. The various types of wounds will also increase the complexity of physical examination at hospital admission and the likelihood of pressure injury development during hospitalization. Therefore, this study's findings are consistent with those reported in the literature.

Systolic blood pressure also affects the development of inpatient pressure injury. Excessively low systolic blood pressure leads to the accumulation of blood in major organs; consequently, the microvascular blood volume and blood vessel elasticity of the skin decrease, which cause not only insufficient perfusion of the skin but also reduction of the skin perfusion pressure. In this case, the amount of oxygen and nutrients supplied to body tissues and cells is low, substantially increasing the rate of pressure injury. Additionally, diseases causing low systolic blood pressure are complex, which includes primary and secondary hypotension. For example, secondary hypotension involves acute and chronic hypotension, which is caused by complex factors. Chronic hypotension is associated with tuberculosis, malignant tumors, and malnutrition; in particular, malnutrition is listed as a key factor in several assessment scales of pressure injury, including the Braden Scale, Norton Scale, Waterlow Scale, and

		Predicted class	
		Pressure injury	Non-pressure injury
Actual class	Pressure injury	<i>TP</i>	<i>FN</i>
	Non-pressure injury	<i>FP</i>	<i>TN</i>

FIGURE 1. Confusion matrix.

Table 3. Variable Importance Calculated by GainRatioAttributeEval

Rank	Variable	Gain Ratio
1	Skin integrity	0.01328
2	Systolic pressure	0.01262
3	Expression ability	0.01169
4	Capillary refill time	0.01093
5	Level of consciousness	0.00987
6	Eye opening (E)	0.00886
7	Level of mobility	0.00860
8	Emotional responses	0.00848
9	Diastolic pressure	0.00801
10	Skin properties	0.00697
11	Color at the peripheral limbs	0.00588
12	Pulse rate	0.00342

Gosnell Scale. Thus, low systolic blood pressure is a risk factor for pressure injury and should be treated as a warning sign. Determining whether the inpatient's malnutrition is caused by his or her disease is a crucial process to reduce the development of pressure injury resulting from diseases.

Expression ability was closely related to the level of consciousness. Restricted activity was observed in inpatients with low consciousness or those in an unstable state of acute illness; this can cause prolonged pressure on the skin then increase the rate of pressure injury. Therefore, the consciousness and physical and mental stress of an inpatient with poor expression ability should be evaluated and observed to avoid neglecting or misidentifying the cause of pressure injury.³⁷ This finding is consistent with those reported in the literature.³⁴

Experimental Results for Prediction Models

Table 4 shows the results of both training and validation sets. As for the training sets, the results indicated that for the prediction model built using DT, LGR, and RF, the average precision is 0.969, 0.799, and 0.998; the average recall is

1.000, 0.827, and 1.000; the average F1-measure is 0.984, 0.813, and 0.999; and the average AUC is 0.987, 0.876, and 1.000, respectively. These results indicate that the prediction model built using RF has the most favorable classification performance. It was also noticed that RFs have nearly perfect prediction performance in all training sets; this is due to all the training sets being resampled by bootstrap method to overcome the class imbalance problem. The distribution of classes in training sets is no longer the same as the original dataset, which may overestimate model performance.

Because all the three classifiers have excellent prediction performance (AUC > 0.8) in the training phase, their use was furthered in the validation phase. The results showed that for the prediction model built using DT, LGR, and RF, the precision is 0.042, 0.041, and 0.044; the recall is 0.802, 0.698, and 0.872; the F1-measure is 0.080, 0.078, and 0.084; and the average AUC is 0.771, 0.767, and 0.864, respectively. As expected, RF performed the best, followed by DT and LGR. Although the recall and the AUC of the RF classifier remained stable (>0.8), the precision rate and the F1-measure of RF classifier decreased significantly. The result is reasonable because the training sets were resampled at a 1:1 ratio and the classifier is forced to treat both classes equally during the supervised learning process. In other words, the well-trained prediction models give higher probability to the minority class (pressure injury). In contrast, the validation set holds original distribution of the dataset, which is highly imbalanced. While using these prediction models on the validation set, there is a strong possibility that they will predict unseen samples as pressure injury and thus generate more FP cases. Even so, the RF classifier retains excellent recall rate (0.872), meaning that 87.2% of pressure injury inpatients can be identified accurately.

DISCUSSION

The results demonstrated that inpatient pressure injury was significantly influenced by 36 of 54 variables in this study. Some critical variables, including skin integrity, systolic blood pressure change, and expression ability, are consistent with studies reported in the literature.^{2,3,7,25,26} However, some

Table 4. Evaluation Results for DT, LGR, and RF Classifiers

Dataset	Methods	Pre	Rec	F1	Sen	Spe	AUC
Training set (10-fold)	DT	0.969	1.000	0.984	1.000	0.968	0.987
	LGR	0.799	0.827	0.813	0.827	0.792	0.876
	RF	0.998	1.000	0.999	1.000	0.998	1.000
Validation set	DT	0.042	0.802	0.080	0.802	0.731	0.771
	LGR	0.041	0.698	0.078	0.698	0.761	0.767
	RF	0.044	0.872	0.084	0.872	0.721	0.864

The values in bold indicate the best outcomes of three methods in each evaluation measurement.
Abbreviations: Pre, precision; Rec, recall; F1, F1-measure; Sen, sensitivity; Spe, specificity; AUC, area under the curve.

unique variables, such as betel nut chewing and alcohol drinking, are = habits among Taiwanese patients and are not common in other countries. Considering distinct variables by culture and countries makes this study an interesting work and a tailor-made healthcare suggestion. With the important variables of this study, nurses can reduce the evaluation time from many indexes, instantly know the high-risk pressure injury evaluation score, give more appropriate nursing care to reduce the occurrence of adverse events, and improve the overall quality of care. These measures can also be used to justify additional budget funding for more support staff in higher-workload clinical environments. Administrators will need these data to convince executives to increase budgets for improved care on their units.

This study made the following three major contributions academically and practically:

1. A simple but effective prediction model can be constructed using data from a hospital database, and the model can then be used as a screening tool for high-risk groups of pressure injury. A comprehensive clinical early warning system can thus be developed based on this model for preventing pressure injury in nursing care practice.
2. Exploring the expression of risk factors for inpatient pressure injury enables clinical nurses to obtain a comprehensive understanding of the characteristics of inpatients with pressure injury. Such information can reduce nursing workload, length of hospital stay of the inpatient, and medical costs associated with pressure injury, as well as improve the quality and safety of hospitalization.
3. Given that domestic studies have rarely applied data mining in inpatient pressure injury research, the information provided by this study is helpful for future research. In addition, this study offers hospitals in Taiwan clear and concrete principles for monitoring and managing their quality indicators for inpatient pressure injury.

CONCLUSION

Inpatient pressure injury may occur in patients from any medical ward. There have been no universal risk assessment criteria available domestically in Taiwan or internationally for pressure injury at the time of writing. Risk factors for pressure injury are multiple and complex. Nurses are required to perform a risk assessment of pressure injury on patients before their admission to hospitals. Nonetheless, because of the nursing shortage and a lack of accuracy of existing pressure injury assessment scales, early detection and prevention of pressure injury are difficult. Thus, inpatient pressure injury is a crucial issue in quality management.

This study used the data mining technique to build prediction models for inpatient pressure injury and conducted an empirical analysis to understand the critical risk factors for inpatient pressure injury. The present study discovered 10 significant risk factors for inpatient pressure injury, which have not been included in the existing four major pressure injury risk assessment scales. This reveals that data mining techniques can extract various types of hidden information, which overcomes the shortcomings of typical statistical analysis. The prediction models developed in this study can help nurses to establish suitable assessment systems for managing risk factors of pressure injury to reduce their workload. Therefore, nurses can provide satisfactory medical care for patients, effectively improving the quality of medical care.

This study is not without limitations; first, the research results may not be inferred to other hospitals due to the differences in medical resources and medical professionals of different institutions, so the information content may vary depending on the size of the hospital. Second, the hospital database used in this study has only 1.5% pressure ulcer cases of all inpatients records within the research period. The low case rate is different from other studies, indicating that the inpatient pressure injury rate is about 0.4% to 38%. It is possible that nursing staff may neglect to report the existence of pressure injury because of excessive workload; therefore, the validity of research results may become the limitation of this study.

In the future, the inpatient pressure injury assessment model developed in this study can be incorporated into clinical decision support systems to convert patients' records and data into valuable information. Nurses should provide enhanced care and case management for high-risk patients with pressure injury identified by the model and establish a comprehensive nursing plan according to the assessment of what these patients need from nursing care. They should also monitor the patients' risk of pressure injury continuously and introduce effective nursing measures proactively for prevention, such as educating patients and their families with appropriate healthcare suggestions and implementing evidence-based medical preventive practices, to reduce the risk of inpatient pressure injury, improve inpatient care quality, and ultimately lower hospital medical expenses.

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